



Rainfall, population density and voter turnout

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ABSTRACT

We create a dataset linking rainfall amounts to constituency-level election data for Irish general elections over the period 1989–2016. Rainfall is shown to significantly reduce voter turnout. The marginal effect of rainfall on turnout is greater in densely populated constituencies, where a rainy day decreases turnout by as much as three percentage points (or five percent). Using a theoretical framework based on a rational voting model, we propose two possible explanations for this effect. Firstly, if rural voters have higher civic duty than urban voters, they may be immune to rain on election day. Secondly, mode of transport may play a role. Urban voters are more likely to travel on foot or bicycle, whereas rural voters typically travel by car. Therefore, the cost to voting associated with rainfall may be higher in urban areas. Constituency-level data on mode of transport from 1997 to 2016 provides some empirical support for this hypothesis.

1. Introduction

Voter participation in the electoral process is key to a functioning representative democracy, with consequences for electoral outcomes, public policy, income inequality and economic growth (Fowler, 2013; Mueller and Stratmann, 2003). There exists a vast literature examining the determinants of voter turnout. These studies generally fall into one of two categories - individual level or aggregate level research. At the individual level, age, education, residential mobility, media exposure and political interest are found to impact a person's likelihood of voting (see Smets and van Ham (2013) for a review of the literature). At an aggregate level, a large population, a very one-sided election and a majority electoral system make it less likely that a single vote will affect the outcome, thereby depressing turnout (see Geys (2006) and Cancela and Geys (2016) for reviews). Population concentration can also affect turnout due to weaker voting traditions in high density urban areas (Funk, 2010; Kavanagh et al., 2004).

Weather conditions on the day of voting often receive a great deal of attention, and generate much discussion, among political analysts, the media and the general public. However, while other factors have been studied extensively, relatively little is known about the effect of weather on voter turnout. There are some notable exceptions, most of which indicate that increased rainfall is associated with lower turnout (Leslie and Ari, 2018; Arnold, 2018; Arnold and Freier, 2016; Artes, 2014; Gomez et al., 2007; Ben Lakhdar and Dubois, 2006). Persson et al.

(2014), on the other hand, find no effect for Swedish elections.

In addition to providing further evidence on the effect of rainfall on turnout, the distinguishing contribution of our work is to link rainfall with population density. By examining the interaction of these two variables, we can assess whether rainfall has stronger effects on voter turnout in high density constituencies. In Section 2 we develop a framework, based on a rational voter model, to understand why these interaction effects may exist. In summary, there are two reasons. The first relates to urban voters having weaker voting norms and therefore being more dissuaded by bad weather than rural voters. This hypothesis has some existing empirical support in earlier work by Knack (1994), who finds that voters in the US with a strong sense of civic duty are "immune to costs associated with bad weather". We also suggest a potential role for different modes of transport. People in urban areas are more likely to travel on foot or by bicycle, whereas rural voters are more likely to travel by car. Therefore, the effort of travelling to the polling station on a rainy day may be higher for some voters in high density constituencies.

Ireland provides an ideal setting for this type of study for a number of reasons. Firstly, Ireland receives more rain on average in a year than any of the other countries in the aforementioned studies.¹ Secondly, while there is day-to-day variation in rainfall amounts across regions in Ireland, average rainfall over the long-term is evenly distributed throughout the country. This provides us with variation in election day rainfall across constituencies, as well as variation within constituencies

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¹ Yearly averages in mm.: Ireland - 1118, France - 867, United States - 715, Germany - 700, Spain - 636, Sweden - 624. Source: Food and Agriculture Organization.

over time. Finally, Irish constituencies encompass a wide range of population densities. The regional population per square kilometre in 2016 ranged from a high of approximately 7000, to a low of 30.

Our dataset matches election day rainfall to voter turnout in 291 constituency level elections in Ireland over the period 1989 to 2016.² A rainy day, defined as 30 mm of rain, is found to reduce turnout by two percentage points. Voter turnout is found to be lower in areas with high population densities. Moreover, the marginal effect of rainfall on voter turnout increases as population density increases. A rainy day in a high density constituency may reduce turnout by as much as three percentage points (or five percent).

We also construct constituency-level statistics for a number of other explanatory variables. The average age in a constituency is shown to play a strong role. A one year increase in average age is associated with an increase in turnout of between one and two percentage points. The evidence also indicates that an increase in marriage rates and educational attainment leads to higher turnout. We create a measure of electoral uncertainty and show that greater uncertainty is associated with increased voter participation. Finally, we examine whether interaction effects exist between rainfall and these additional explanatory variables. We find that rain is associated with lower turnout only in areas with young populations. As age can be viewed as a proxy for civic duty (Lacy and Burden, 1999), this result is consistent with the idea that voters with high sense of civic engagement are more immune to bad weather.

The remainder of the paper is organised as follows. In Section 2 we develop a theoretical framework to understand why the marginal effect of rainfall on voter turnout may be conditional on population density. Section 3 describes the newly created dataset and gives an overview of the Irish electoral system. Section 4 presents some descriptive statistics on voter turnout and rainfall. In Section 5 we describe our estimation methodology and in Section 6 we discuss our results. Section 7 concludes.

2. Theoretical framework

We take a rational voting model, such as that used in Lacy and Burden (1999), and extend it to provide a theoretical framework for understanding why rainfall may interact with population density to affect voter turnout. Individual i 's utility for voting is given by,

$$U_i = P_i B_i - C_i + D_i \quad (1)$$

where P is the probability of casting a decisive vote, B is the benefit from having one's preferred candidate elected, C is the cost of voting and D represents an individual's civic duty, which can be viewed as the consumption value of voting. Election day rainfall is contained within C . Separating rainfall, denoted R , and dropping subscripts for convenience, gives,

$$U = PB - R - C + D \quad (2)$$

Treating PB as fixed, the marginal effects of the other factors are represented by the parameters α_1 , α_2 and α_3 .

$$U = \overline{PB} - \alpha_1 R - \alpha_2 C + \alpha_3 D \quad (3)$$

Therefore $\frac{\partial U}{\partial R} = -\alpha_1$ represents the marginal reduction in voter utility associated with increased rainfall and $\frac{\partial U}{\partial D} = \alpha_3$ indicates the marginal positive impact on utility associated with a higher sense of civic duty.

Civic duty may interact with rainfall. For example, bad weather on election day may have more of an impact on those with a weak sense of civic duty. This hypothesis has some empirical support in previous work

by Knack (1994) who suggests that voters with a greater civic duty are immune to bad weather. Incorporating this interaction gives,

$$U = \overline{PB} - \beta RD^{-1} - \alpha_2 C \quad (4)$$

Note that we are using the inverse of civic duty, D^{-1} . This implies that, $\frac{\partial U}{\partial R}|_{D=D_L} < \frac{\partial U}{\partial R}|_{D=D_H}$, i.e., the negative marginal impact of increased rainfall on utility $-\beta D^{-1}$, is greater for those with low civic duty, D_L , compared to those with high civic duty, D_H . Given that civic duty, on average, tends to be higher in rural areas (see, e.g., Funk, 2010; Kavanagh et al., 2004), this implies that rainfall will have more of an effect on voters in high density areas, a hypothesis which is at the core of our empirical analysis.

The means by which a person travels to the polling station may also represent a cost to voting. Let W be a binary indicator of whether a person walks to the polling station. This, on its own, may affect voter utility. However, it is likely that W will also interact with R . For example, the cost of walking to the polling station will vary with rainfall, with more rainfall imposing a greater cost. We separate W from other costs, C , and interact with rainfall to get,

$$U = \overline{PB} - \beta RD^{-1} - \lambda RW - \alpha_2 C \quad (5)$$

The effect of rain on voter utility is therefore given by, $\frac{\partial U}{\partial R} = -\beta D^{-1} - \lambda W$. The impact increases for those who walk to the polling station and decreases with civic duty. As we show later in the paper, individuals in high density (urban) areas are more likely to travel by foot, while those in low density (rural) areas are more likely to travel by car. Therefore population density may interact with rainfall for reasons relating to mode of transport, in addition to reasons relating to a sense of civic duty. Taken together, this implies that $\frac{\partial U}{\partial R}|_{\text{high density}} < \frac{\partial U}{\partial R}|_{\text{low density}}$, i.e., the negative impact of rainfall on voter utility is greater in high density regions.

Note that our theoretical framework examines voter utility as opposed to focusing directly on voter participation. However, both concepts are clearly interrelated, as factors that are deleterious to utility will reduce the probability of voting. For example, defining V as a binary indicator of whether the person decides to vote, we could frame the participation decision in relation to equation (1) as follows: $V = 1$ if $PB + D > C$, meaning that the individual votes if the benefits of doing so exceed the costs. The theoretical predictions of our framework therefore indicate that the marginal effect of rainfall on voter turnout will vary with population density. It is precisely this relationship which we test. We also attempt to test the prediction that mode of transport plays a role. However, precisely disentangling the effects of civic duty and mode of transport is difficult.

Another potential link between population density and turnout relates to the distance from the polling station. Rural voters will typically have a longer distance to travel to the polling station, which could impose a higher cost of voting and therefore depress turnout (Gimpel and Schuknecht, 2003; Haspel; Knotts, 2005). However, the relationship is not linear. As noted by Gimpel and Schuknecht (2003), the heaviest burden occurs in the middle range of distance (2–5 miles), whereas for rural voters with longer distances to travel (6–10 miles), the route is direct and unimpeded and consequently turnout is higher. Related to this point, it is not clear that rainfall should interact with distance for these rural voters living far from the polling station. With such a distance to travel, these voters will likely use their car. As such, whether or not it is raining on election day may be of less importance to rural voters. Conversely, voters in urban areas who may be closer to the polling station, but less likely to use their car, may be more affected by election day rainfall.

One could also consider the possibility of other interactions. However, while factors such as the closeness of the election or a person's education level may impact the probability of voting, their impact may be felt before election day and are therefore unaffected by weather. For example, education may impact whether a person is a "voter" or "non-voter". Therefore, the sorting between the two voter types may have

² Due to a combination of data restrictions relating to electoral statistics, census data and disaggregated rainfall data, we cannot go beyond 1989 as our starting point.

taken place well in advance of election day. Nonetheless, we empirically test for the presence of other types of interactions.

3. Electoral system and data construction

Ireland is a parliamentary democracy with two Houses of Parliament. The upper house is known as Seanad Éireann and members of this house are either nominated by the sitting prime minister or elected by specialist panels. The lower house is known as Dáil Éireann and is made up of directly elected members of parliament, known as Teachta Dála, or simply TDs. The lower house is responsible for introducing legislation and making policy decisions. While the upper house can review and delay legislation, it cannot stop the legislation from being enacted.

In this paper, we focus on elections to the lower house, which are held at least once every five years. A system of proportional representation with a single transferable vote (PR-STV) is used to elect between 3 and 5 TDs in each of the 40 constituencies. PR-STV allows voters to rank candidates on the ballot paper in order of their preferences. For example, a voter places a 1 beside her most preferred candidate, a 2 beside her second preference, and so on. An election quota is calculated by dividing the total number of valid votes by one more than the number of seats to be filled, ignoring any remainder and then adding one.³ A candidate is elected once her votes surpass the quota. Only one of the voter's preferences is active at any one time. For example, a vote stays with the highest preference candidate until that candidate gets elected or eliminated, at which point it transfers to the voter's next highest preference candidate that is still in the running. Following a count, if no candidate has enough votes to secure election, the least voted candidate is eliminated and her votes are transferred. A subsequent count then takes place and this process continues until all seats have been filled.

We create a dataset which matches rainfall amounts on election days, in millimetres, to each electoral constituency. Our data spans the period from 1989 to 2016, comprising 291 constituency races from seven general elections. Rainfall data is collected from the historical database of Ireland's national meteorological service, Met Éireann. This database provides daily weather information from 511 weather stations around the country. The exact location of these stations is represented in Fig. A.1 in Appendix. As can be observed, the stations provide wide coverage of the whole country.

To measure the amount of rain in a given constituency, we first take the coordinates of each weather station, as provided by Met Éireann, and allocate each weather station to an electoral division, which is the smallest legally defined administrative area. We then map the electoral divisions into their electoral constituencies as defined by the Electoral (Amendment) Act corresponding to the appropriate election. In a small number of cases, several stations are located in very close proximity within the same electoral division. Where this occurs, we take the average of all the stations situated within the same electoral division. Finally, we take the average rainfall values from weather stations within each constituency to produce our rainfall measure.

In line with Gomez et al. (2007), we also create an alternative measure for rainfall that accounts for the possibility that voters in some areas may be more accustomed to rain than in other areas. We calculate the average daily rainfall for the election month and use the election day deviation from that average. For example, the weather station in Dublin Airport collected 10.2 mm of rain on the election day of June 6th, 1997. The daily average for the month of June for the available sample on that station is 2.0 mm of rain; therefore, the value of our alternative rainfall measure for the Dublin Airport station is 8.2 mm.

As an additional robustness test, we construct a second alternative measure of rainfall that takes into account the population distribution within the constituency. Within a single constituency, some weather

stations can be located in areas where more voters live, whereas others can be located in more sparsely populated areas. By simply averaging the two, they will have the same weight in determining the overall level of rainfall at the constituency level. To account for this, we create another measure of rainfall which is an average of all weather stations within the constituency, weighted by the population distribution within that constituency. Therefore, this approach assigns more weight to weather stations in more populated areas. As we show, our results are robust to both of these alternative rainfall measures.

We also create a measure of election uncertainty, applicable to the PR-STV system. Measuring uncertainty, or election closeness, in PR-STV is complicated by the multi-seat nature of elections coupled with the system of vote transfers.⁴ We attempt to proxy uncertainty using the variable, $Counts_{i,t}$, which is calculated by dividing the number of counts by the number of candidates. As outlined at the beginning of this Section, a candidate is elected once they meet the election quota. If no candidate is elected, then the lowest voted candidate is eliminated, their votes are transferred, and a subsequent count takes place. Therefore, if a large number of counts are required to fill all seats, this suggests that there was a closely fought contest for at least one of the available seats. As an example, consider the 2007 election in Carlow Kilkenny, where 11 candidates were vying for five seats. A total of nine counts were required, during which all of the losing candidates were sequentially eliminated, before all seats were filled. The margin of victory for the final seat was close, with two candidates receiving a similar number of votes. A problem of using counts to measure election closeness or uncertainty is that it is correlated with the number of candidates running. A large number of candidates would naturally disperse the allocation of votes, increasing the number of counts necessary to push a given candidate over the electoral quota. This is why we divide the number of counts by the number of candidates.

Our measure of population density is defined as the population per square kilometre. We also use a variable which indicates the size of the electorate, i.e., the number of registered voters within the constituency. Finally, we explore the role of education level, marital status and age on voter turnout. These three variables are taken from census data. However, due to limited availability of disaggregated census data, they are not available for all years. As such, we present our baseline specification which includes all of our rainfall and turnout data, followed by a supplementary specification which includes these additional variables for a reduced subsample.

Our dependent variable, voter turnout, is the number of votes cast within a constituency as a percentage of the total electorate within that region. Turnout data is taken from irelandelection.com, which is a repository of Irish election data statistics. Nealon's Guide election publications are used to supplement the data in cases of missing information, as well as to verify the data on irelandelection.com.

4. Descriptive statistics

Fig. 1 shows a scatterplot of voter turnout and rainfall for each of the constituency-level observations. There is a negative relationship which indicates that greater rain on election day is associated with lower turnout rates. In Table 1, we collect these observations at a more aggregate level to show national voter turnout rates and the associated average rainfall amounts for each of the seven general elections. The three election days with the greatest national average rainfall, occurred in 2002 (21 mm), 1997 (9 mm) and 2016 (8 mm), were also the three years with the lowest national average turnout rates in our sample.

There is substantial election day variation in rainfall amounts across

³ For example, in a constituency with four seats and 10,000 valid votes, the election quota is $(10,000/(4 + 1)) + 1 = 2001$.

⁴ In related work, Dubois and Leprince (2017) create a measure of election closeness for French open-list elections. Lists are ranked based on the number of seats won. The closeness measure is the difference in the share of seats obtained in the first and second ranked lists.

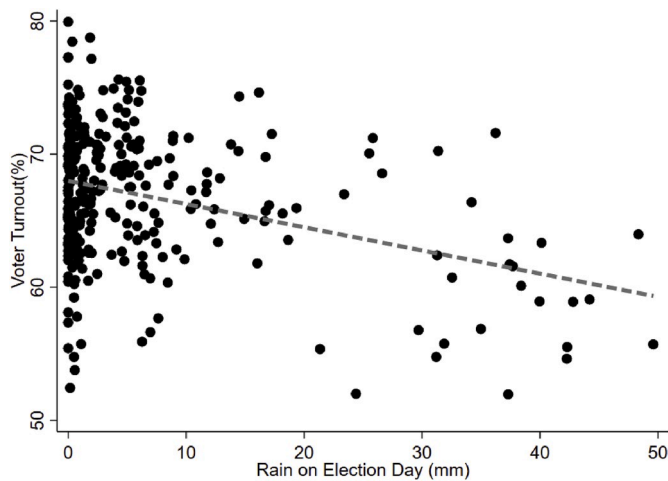


Fig. 1. Rain in election day (mm.) and turnout (%) by constituency, 1989–2016. Dashed line represents the fitted least squares regression line.

Table 1
Average rainfall and voter turnout.

Year	Rainfall (mm)	Turnout (%)
1989	0.09 min = 0, max = 1	69.0 min = 57, max = 80
1992	3.04 min = 0, max = 10	68.7 min = 59, max = 76
1997	9.46 min = 2, max = 20	66.1 min = 56, max = 75
2002	20.96 min = 0, max = 44	62.6 min = 52, max = 73
2007	0.89 min = 0, max = 6	67.0 min = 54, max = 78
2011	2.51 min = 0, max = 6	69.8 min = 61, max = 79
2016	8.24 min = 0, max = 49	64.8 min = 52, max = 72

Notes: Average election day rainfall for the entire country, in mm per 24 h is shown along with maximum and minimum values from weather stations across the country. National average turnout is shown along with minimum and maximum values taken from individual constituencies.

constituencies. This is of particular importance for our study as we are estimating the effect of rainfall on turnout using constituency-level data. Fig. 2 shows boxplots of election day rainfall for each election year. It is clear that, even in the rainiest of election days, there is substantial

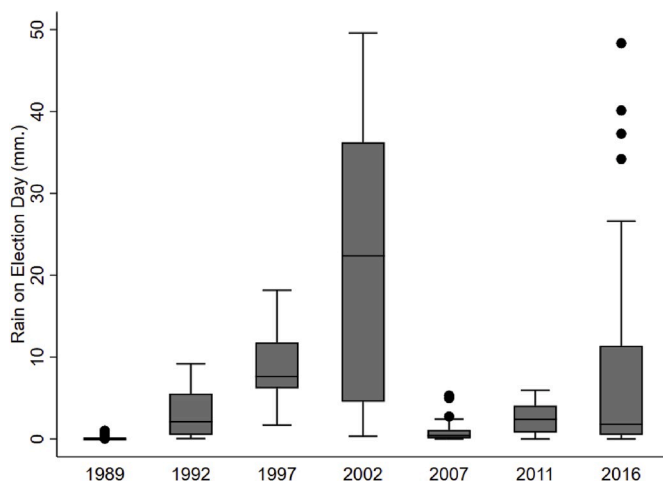


Fig. 2. Boxplot by election year of rain in election day (mm.) by constituency.

variation. In 2002, for example, even with a national average over 20 mm, some constituencies received virtually no rain on election day. On the other hand, in 2016 we can observe how the national median was very low, but a proportion of constituencies were heavily affected by rain.

To further show the degree of variation in rainfall and, more importantly, its geographical variation, we map rainfall amounts across the country for the two rainiest election days (Fig. 3). In 2002, the east, midlands and northwest experienced particularly heavy rain, while the west and southwest remained reasonably dry. In 2016, the heaviest rainfall was confined to the southern parts of the country. In 1997, the third rainiest election, which is not mapped here, heavy rain was concentrated on the coastal areas.

Recently, Lind (2019) has raised concerns about the potential existence of spurious relationships between rainfall and other spatially correlated outcomes. The geographic dispersion of the exposure to the variable of interest, i.e., rainfall, documented in our study should ensure that we are capturing the actual effect of rainfall on participation and not some other unobserved regional characteristic. Moreover, we also employ a constituency fixed effects specification, which will control for any potential spatial trends.

The average turnout for all years in the sample is 66.88%. To put these numbers in an international context, the OECD average for parliamentary elections for the 1989–2017 period is 71.55% (69.46% excluding countries with compulsory voting).⁵ By constituencies, the lowest value was registered in the Dublin South Central constituency during the 2002 election, with a turnout of just 51.96%, and the highest was registered in the Cork North West constituency during the 1989 election, with a 79.94% turnout.

5. Methodology

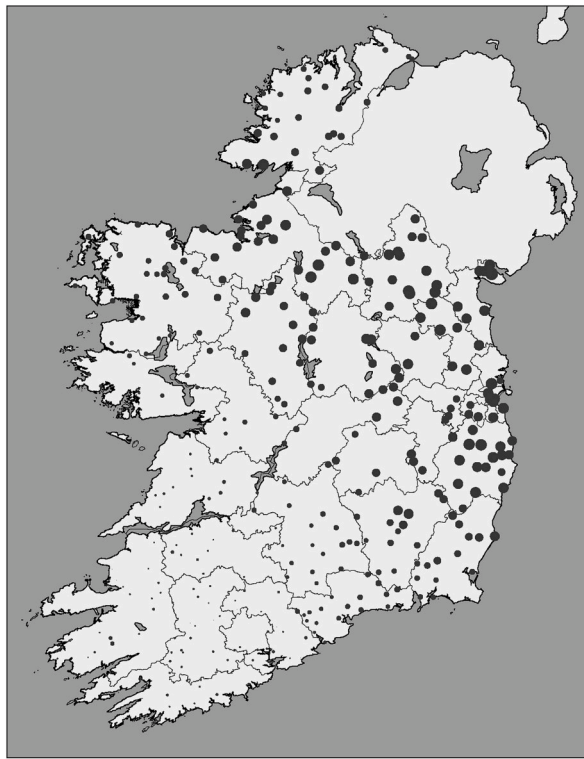
We estimate the effect of rainfall on voter turnout by estimating the following regression,

$$T_{i,t} = \alpha + \beta_1 \text{Rain}_{i,t} + \beta_2 \text{PD}_{i,t} + \beta_3 \text{Counts}_{i,t} + \beta_4 \text{Electorate}_{i,t} + \delta_t + \varepsilon_{i,t} \quad (6)$$

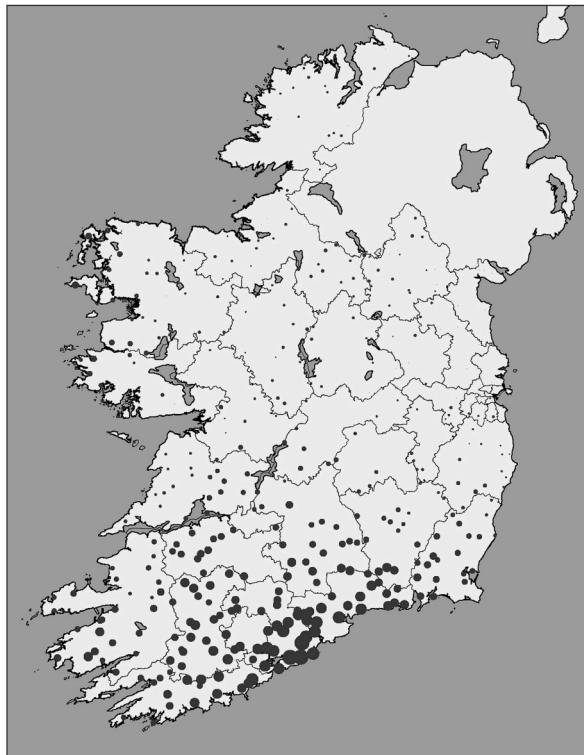
Our dependent variable, $T_{i,t}$, denotes voter turnout in constituency i at time t . Our main variable of interest, $\text{Rain}_{i,t}$, is election day rainfall in millimetres in constituency i at time t . Additional independent variables include $\text{PD}_{i,t}$, which is the log of population density (population per square kilometre) and $\text{Electorate}_{i,t}$, which is the log of the size of the electorate. There may be concerns that these two variables are correlated, as constituencies with higher population density may be expected to have a larger electorate. However, the most densely populated constituencies tend to also be the smallest in geographical terms, and as such density and population are not strongly correlated. Moreover, we experiment with dropping either variable and observe that the estimate for the other variable remains virtually unchanged. $\text{Counts}_{i,t}$ is our proxy for election uncertainty based on the number of counts required to fill all seats, which was explained in detail in Section 2. We also include time fixed effects, δ_t , which control for any election specific shocks that impacted all constituencies in a given year. This could include, for example, the day of the week the election was held, or the national economic conditions prevailing at the time.

Note that our dataset is an unbalanced panel which follows constituencies over time for a maximum of seven elections. Therefore, in addition to pooled OLS (equation (1)), we employ panel estimation techniques. The presence of unobservable, time-invariant, constituency-specific effects could bias our estimates in equation (6) if these unobservable effects are correlated with our independent variables. This can be addressed by employing the fixed-effects regression shown below (equation (7)). The difference between equations (6) and (7) is the term

⁵ Data from the International Institute for Democracy and Electoral Assistance (International IDEA) Voter Turnout Database.



2002



2016

Fig. 3. Rain on election day. Every dot represents a weather station and the size of the dot represents the amount of rain collected on every station. Black lines represent the 2016 constituency borders.

θ_i , which is the constituency specific fixed effect.

$$T_{i,t} = \alpha + \beta_1 \text{Rain}_{i,t} + \beta_2 \text{PD}_{i,t} + \beta_3 \text{Counts}_{i,t} + \beta_4 \text{Electorate}_{i,t} + \delta_i + \theta_i + \varepsilon_{i,t} \quad (7)$$

A random effects estimator may be used if the unobservable effects are not correlated with the independent variables. As a guide to which is the more appropriate estimator, we implement a Hausman specification test (see [Hausman, 1978](#)). While the Hausman test indicates that a random effects estimator may be used, we report results from the random effects, fixed effects and pooled OLS estimators for robustness.

We test for interaction effects between rainfall and population density. This can be done by estimating the regressions shown above with the inclusion of an additional interaction term between density and rainfall,

$$T_{i,t} = \alpha + \beta_1 \text{Rain}_{i,t} + \beta_2 \text{PD}_{i,t} + \beta_3 \text{Rain} * \text{PD}_{i,t} + X'_{i,t} \varphi + \varepsilon_{i,t} \quad (8)$$

The coefficient β_3 is the estimate of the interaction effect, which tells us whether the effect of rainfall on turnout varies with population density. For brevity, $X_{i,t}$ is used in equation (8) to denote the vector of additional independent variables. To ensure current best practice in the implementation and interpretation of interaction effects, we follow the guidance of [Hainmueller et al. \(2018\)](#) and [Brambor et al. \(2005\)](#). Instead of looking at the statistical significance of the estimate β_3 in isolation, we plot the marginal effects of rainfall on voter turnout across the full range of values of the interacting variable (population density).⁶ For example, one could observe β_3 as being not statistically significant, yet the marginal plots could reveal that it is statistically significant for a specific range of the interacting variable.

While equation (8) imposes linearity on the interaction effect, [Hainmueller et al. \(2018\)](#) propose a binning estimator which allows for non-linearities. This involves breaking the interacting variable into three discrete bins corresponding to the terciles of that variable. More formally, denoting population density as PD , we create the following three bins,

$$B_1 = \begin{cases} 1 & PD < \delta_{1/3} \\ 0 & \text{otherwise} \end{cases} \quad B_2 = \begin{cases} 1 & PD \in [\delta_{1/3}, \delta_{2/3}] \\ 0 & \text{otherwise} \end{cases} \quad B_3 = \begin{cases} 1 & PD \geq \delta_{2/3} \\ 0 & \text{otherwise} \end{cases} \quad (9)$$

where $\delta_{1/3}$ and $\delta_{2/3}$ are the first and third terciles of the population density variable. We use the median population density within each bin as evaluation points to estimate the marginal effect of rainfall on turnout. Denoting the three evaluation points as pd_1 , pd_2 and pd_3 , we estimate the following regression (omitting subscripts for convenience),

$$T = \sum_{j=1}^3 \left\{ \alpha_j + \beta_j^1 \text{Rain} + \beta_j^2 (PD - pd_j) + \beta_j^3 (PD - pd_j) \text{Rain} \right\} B_j + X' \varphi + \varepsilon \quad (10)$$

Given that $(PD - pd_j) = 0$ at the evaluation point pd_j , the marginal effect of rainfall on voter turnout at each of the three evaluation points is given by β_1^1 , β_2^1 and β_3^1 . We can view these three estimates as the marginal effect of rainfall on voter turnout for low density constituencies, mid density constituencies and high density constituencies.

Finally, [Hainmueller et al. \(2018\)](#) highlight a common issue relating to a lack of common support when interpreting interaction effects. When looking at the marginal effects across a range of values of the interacting variable, if there are not enough observations across the range of values, then the estimates within such ranges are reliant on extrapolation. To verify common support, our graphs include a histogram showing the

⁶ As both population density and rainfall form part of the interaction, either one could be called the interacting variable. For clarity, and given rainfall is our main variable of interest, we refer to population density as the interacting variable.

distribution of the population density variable.

6. Results

Table 2 shows results from our three estimators: pooled OLS, random effects and fixed effects.⁷ We also report results from our alternative rainfall measure which uses the deviation from the monthly average rainfall amounts, as well as the rainfall measure which takes a weighted average of the weather stations based on the population distribution within the constituency. The results using the alternative rainfall measures are similar to our baseline estimates for all specifications. For brevity, we just report the OLS results for the alternative rainfall measures. In all of the tables that follow, we denote the monthly deviation measure with a column heading “Rain 2” and the weighted average measure with a column heading “Rain 3”.

The estimated effect of rainfall on voter turnout is similar across all specifications. A one millimetre increase in election day rainfall is associated with a reduction in voter turnout of approximately 0.06 percentage points. To put this result into context, we look at Met Éireann’s weather warnings. Ordered from lowest to highest in severity, the three warnings are yellow, orange and red. Yellow indicates that, while not unusual weather, there may be localised danger. Orange weather is described as infrequent and dangerous or disruptive. Red weather is rare and is classified as extremely dangerous.⁸ Rainfall of 70 mm or more in 24 h triggers a red warning. Orange warnings relate to rainfall of 50 mm–70 mm, while yellow warnings relate to 30 mm–50 mm. Rainfall below 30 mm does not require any warning. Therefore, it is reasonable to define a moderately rainy day as rainfall of 30 mm in 24 h. Our estimates indicate that if an election occurs on this type of rainy day, then voter turnout will be reduced by approximately two percentage points (or three percent).

There is also evidence that turnout decreases as population density increases. In addition, voter turnout is higher when there is more electoral uncertainty. Note that we use an *ex-post* measure of uncertainty. Ideally, one would prefer an *ex-ante* measure. However, these are typically unavailable to researchers. In a review of the literature on voter turnout, Geys (2006) finds that over 70 percent of these types of measures are *ex-post* proxies. The consequence of this is that we cannot rule out some degree of simultaneity bias.

6.1. Rainfall and population density

We examine the interaction between rainfall and population density by estimating equation (8). The results are shown in Table 3. The interaction coefficient is negative and statistically significant across all specifications, indicating that the marginal effect of rainfall on voter turnout increases, i.e., exerts a greater downward influence on turnout, as population density increases.

However, it is not sufficient to view the coefficient on the interaction effect in isolation (see, e.g., Hainmueller et al., 2018; Brambor et al., 2005). Even if the interaction coefficient in a regression is not statistically significant, the marginal effect of the variable of interest on the dependent variable may be significant for a specific range of the interacting variable (i.e., population density). We plot the marginal effects of rainfall on turnout across the population density distribution (Fig. 4).

⁷ The Hausman test fails to reject the null hypothesis that the constituency-level effects are adequately modelled by a random effects model. The chi-square test statistic is 3.89 and the p-value is 0.9522. However, for robustness we also report the fixed effects results.

⁸ Met Éireann uses these warnings as part of their remit to ensure citizen safety. See <https://www.met.ie/weather-warnings> for more details.

We do this for both the pooled OLS and fixed effects model using our baseline measure of rainfall.⁹ To test for non-linearity in the interaction effect, we use our binning estimator to estimate the effect of rainfall on voter turnout for low (L), medium (M) and high (H) density areas.

For low population density constituencies, the marginal effect of rainfall on voter turnout is not statistically significant. However, the marginal effect increases as population density increases. For high density constituencies, a 1 mm increase in rainfall is associated with a 0.1 percentage point decrease in voter turnout. Therefore, a rainy day (30 mm in 24 h) in a high population density constituency reduces voter turnout by approximately three percentage points. Average turnout and total votes in high density constituencies is approximately 63 percent and 45,000 votes respectively. Therefore, a rainy day in this type of constituency, is associated with over 2,000, or five percent, fewer votes.

6.2. Exploring additional variables

We include additional explanatory variables, often cited within the literature as affecting voter turnout, and re-estimate our models for a subset of elections. Using census data, we create constituency-level variables for average age, marriage rates and the percentage of the electorate with a third level education. However, due to limited availability of census data, our analysis is restricted to five general elections from 1997 to 2016. The results are shown in Table 4. Our estimated effect of rainfall on turnout is robust to this alternative specification. As in the previous specifications, the results show that higher population density areas are associated with lower turnout. The interaction effects are also robust to this specification.¹⁰

With regard to the additional independent variables, the results indicate that higher levels of education are associated with increased turnout. A one percentage point increase in the rate of third level education among the electorate is associated with an increase in voter turnout of 0.09 and 0.29 percentage points for the random effects and fixed effects models respectively. There is also evidence that voter turnout increases as marriage rates rise: a one percentage point increase in the marriage rate is associated with a 0.64 and 0.41 percentage point increase in voter turnout in the pooled OLS and random effects models respectively. Where it is found in the literature, the explanation for the effect of marriage on voting, is that even if one of the spouses is politically motivated, this can have spillover effects which influence the other spouse to vote. Finally, a higher average age in a constituency has a very strong effect on turnout across all specifications. A one year increase in average age is associated with an increase in voter turnout of between one and two percentage points.

While we have focused on interacting rainfall with population density, it is possible that other interaction effects exist. We test for the presence of such effects and find that age is the only other variable that interacts with rainfall in a statistically significant way.¹¹ Figure A2 in Appendix shows the marginal effect of rainfall on voter turnout across the age distribution. Rainfall leads to a reduction in turnout only among the youngest constituencies. For older constituencies, the effect is not significant. This is consistent with the hypothesis that age may be a proxy for civic duty (Lacy and Burden, 1999).

As a final robustness test, we estimate a model which simultaneously includes interactions of rainfall with all of the other independent variables to see if the interaction effects for population density and age

⁹ There is no facility to implement random effects with the *interflex* command. However, the results are similar across specifications. The random effects would lie somewhere in between the pooled OLS and fixed effects results.

¹⁰ As the interaction effects are virtually identical to the previous specification, for brevity, we do not include the tables or figures.

¹¹ Estimating equation (8) using age instead of population density as the interacting variable, gives a strongly significant interaction coefficient of 0.019 ($p < 0.01$).

Table 2
Determinants of voter turnout.

VARIABLES	(1)	(2)	(3)	(4)	(5)
	OLS	RE	FE	Rain 2	Rain 3
Rainfall	−0.053** (0.023)	−0.063*** (0.019)	−0.064*** (0.018)	−0.055** (0.023)	−0.061* (0.032)
Pop. density (log)	−1.592*** (0.150)	−1.531*** (0.145)	−0.927 (0.587)	−1.579*** (0.149)	−1.581*** (0.150)
Electorate (log)	−1.360 (1.676)	−1.436 (1.681)	−1.788 (2.058)	−1.354 (1.669)	−1.439 (1.687)
Counts	1.729 (2.116)	3.076*** (1.154)	3.124*** (1.151)	1.714 (2.117)	1.754 (2.092)
Constant	90.292*** (19.067)	89.757*** (19.101)	90.610*** (23.539)	90.032*** (19.006)	91.081*** (19.164)
Year fixed effects	yes	yes	yes	yes	yes
Observations	291	291	291	291	291
R-squared	0.53	0.52	0.61	0.53	0.53
Groups		64	64		

Notes: Column (4), “Rain 2”, shows OLS results from the alternative rainfall measure using deviations from monthly averages. Column (5), “Rain 3”, shows OLS results from the alternative measure which weights weather stations by the population distribution. Standard errors, in parentheses, are clustered at the constituency level.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 3
Determinants of voter turnout (interaction effects).

VARIABLES	(1)	(2)	(3)	(4)	(5)
	OLS	RE	FE	Rain 2	Rain 3
Rainfall	0.036 (0.057)	0.017 (0.052)	0.012 (0.053)	0.037 (0.058)	0.056 (0.075)
Pop. density (log)	−1.490*** (0.152)	−1.447*** (0.139)	−0.869 (0.565)	1.517*** (0.149)	−1.478*** (0.148)
Interaction	−0.016* (0.008)	−0.014* (0.007)	−0.013* (0.008)	−0.016* (0.008)	−0.020* (0.010)
Electorate (log)	−1.447 (1.689)	−1.621 (1.743)	−2.012 (2.175)	−1.472 (1.687)	−1.497 (1.693)
Counts	1.635 (2.086)	2.942*** (1.101)	2.985*** (1.092)	1.617 (2.085)	1.597 (2.064)
Constant	90.822*** (19.163)	91.477*** (19.698)	92.883*** (24.722)	91.146*** (19.158)	91.334*** (19.189)
Observations	291	291	291	291	291
R-squared	0.53	0.53	0.61	0.53	0.53
Groups		64	64		

Notes: Column (4), “Rain 2”, shows OLS results from the alternative rainfall measure using deviations from monthly averages. Column (5), “Rain 3”, shows OLS results from the alternative measure which weights weather stations by the population distribution. Standard errors, in parentheses, are clustered at the constituency level.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

remain. Specifically, we estimate equation (8) and include interactions of rainfall with all of the control variables contained within X'_{it} . The population density and age interactions remain strongly significant. In [Appendix Table A1](#), we show that the interaction effects from this model are robust across our three alternative rainfall measures. The population density interaction effects are actually higher in magnitude and statistical significance compared to our baseline estimates.

6.3. Mode of transport

In [Section 2](#) we suggested that, in addition to civic duty, the mode of transport used by voters to get to the polling station may help explain the interaction between rainfall and population density. If voters in urban areas are more likely to walk, or cycle, to the polling station compared to rural voters, then rainfall may act as a stronger deterrent to voting in high population density constituencies. Ideally, studying this would require information on how voters travel to the polling station on election day in each constituency over time. We do not have such data available to us. However, we do have constituency level statistics on how people travel to work over the time period 1997–2016, which covers five elections. We can use this data to further explore the potential role of mode of transport.

We first examine the relationship between voter turnout and the

percentage of voters that travel to work on foot or by bicycle. This is shown in [Fig. 5](#). We clearly see a negative relationship, that is, voter turnout is lower in constituencies with a high percentage of people travelling on foot or bicycle.

Next, we examine whether there is a relationship between the mode of transport and population density. This is shown in [Fig. 6](#). As expected, we see a strong positive relationship. This indicates that a far greater percentage of individuals travel to work on foot or bicycle in high density areas.

Finally, we test for interaction effects by examining whether the marginal effect of rainfall on voter turnout varies by the percentage of people that travel on foot or by bicycle. [Fig. 7](#) shows that rainfall has a greater negative impact on voter turnout in areas where a high percentage of people walk or use a bicycle. Taken together, these results are consistent with the hypothesis that mode of transport may be one of the channels through which rainfall interacts with population density.

However, caution is called for when interpreting these results. Our measure is simply a proxy for how people may travel to vote. Moreover, we cannot rule out the possibility that this measure is correlated with some other factor which is driving our results. Therefore, while the associations presented in [Figs. 5–7](#) may point towards a possible role for mode of transport, we must refrain from making definitive causal claims.

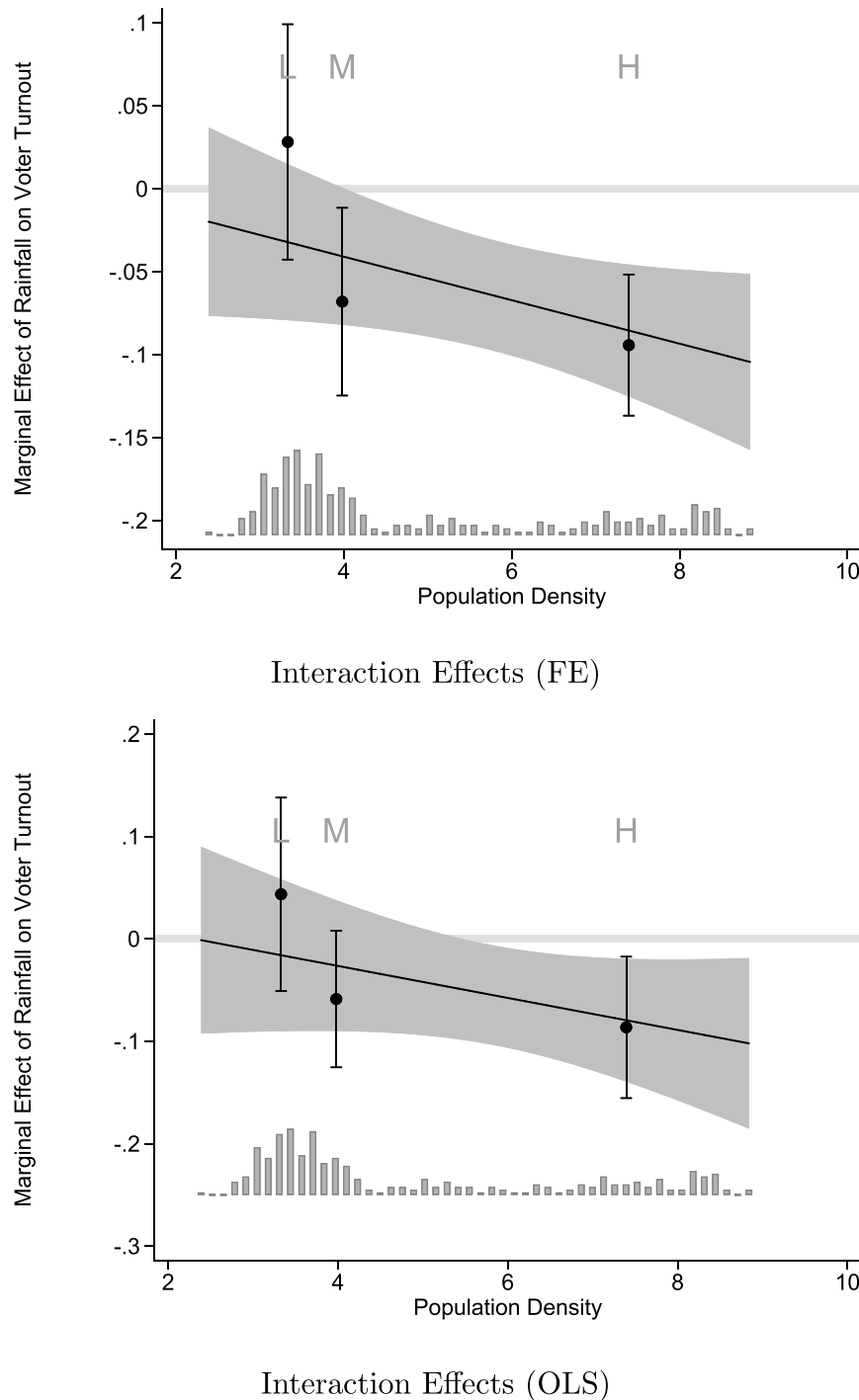


Fig. 4. The marginal effects of rainfall on voter turnout are shown for all values of population density, along with 95 percent confidence intervals. Three point estimates from the binning estimator show the effect of rainfall on voter turnout for low, medium and high population densities.

Given that means of travel to work is not a variable that is thought to directly relate to voter turnout, and given its strong correlation with population density, we chose to examine this effect separately and to caveat the results accordingly, as opposed to including the variable in our main specification. However, it is worth briefly discussing the results from a specification which includes all of the independent variables, including mode of transport, as well as the full range of interactions. Firstly, the age and population density interaction effects remain relatively unchanged from those shown in [Appendix Table A1](#). However, the mode of transport interaction is not statistically significant. However,

we know that mode of transport and population density are strongly correlated. If we omit population density and include all other variables and interactions, then the mode of transport interaction is negative and statistically significant.¹²

¹² The interaction coefficient for mode of transport and rainfall is -0.021 ($p < 0.01$).

Table 4
Determinants of voter turnout (additional variables).

VARIABLES	(1)	(2)	(3)	(4)	(5)
	OLS	RE	FE	Rain 2	Rain3
Rainfall	-0.046** (0.020)	-0.048** (0.019)	-0.047** (0.019)	-0.046** (0.020)	-0.051** (0.024)
Pop.density (log)	-3.881*** (0.223)	-1.342*** (0.325)	-2.251*** (0.686)	-8.70*** (0.223)	-0.876*** (0.221)
Electorate (log)	-1.634 (1.353)	-2.068* (1.131)	-0.985 (1.842)	-1.638 (1.353)	-1.739 (1.363)
Counts	2.368 (1.437)	3.551*** (1.333)	3.958*** (1.007)	2.351 (1.440)	2.384* (1.413)
3rd level education (%)	0.030 (0.023)	0.086* (0.045)	0.289*** (0.073)	0.030 (0.023)	0.030 (0.023)
Married pop. (%)	0.641*** (0.142)	0.405** (0.186)	-0.061 (0.129)	0.641*** (0.141)	0.638*** (0.139)
Average age	0.923*** (0.212)	1.118*** (0.217)	2.036*** (0.184)	0.919*** (0.211)	0.923*** (0.213)
Constant	31.722* (16.127)	39.170*** (12.896)	14.529 (18.831)	31.737* (16.077)	32.845** (16.143)
Year fixed effects	yes	yes	yes	yes	yes
Observations	209	209	209	209	209
R-squared	0.73	0.70	0.75	0.73	0.73
Groups	60	60			

Notes: Column (4), "Rain 2", shows OLS results from the alternative rainfall measure using deviations from monthly averages. Column (5), "Rain 3", shows OLS results from the alternative measure which weights weather stations by the population distribution. Standard errors, in parentheses, are clustered at the constituency level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

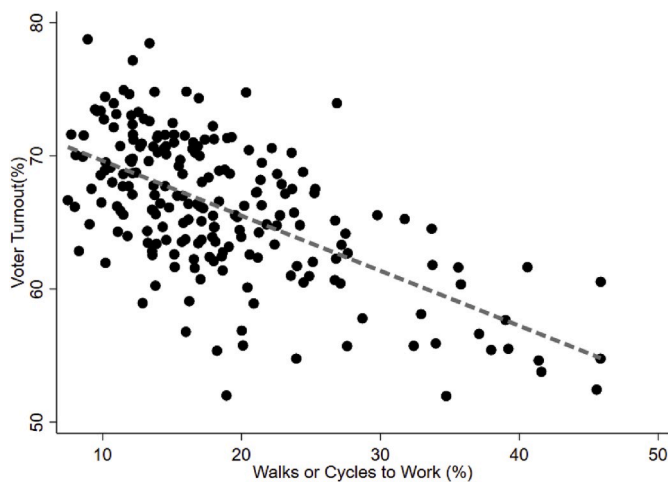


Fig. 5. Voter turnout (%) and the percentage of people that travel to work on foot or bicycle by constituency, 1997–2016. Dashed line represents the fitted least squares regression line.

7. Conclusion

A vast literature exists examining the determinants of voter turnout. However, little attention is given to measuring the effect of adverse weather conditions on turnout. Using a newly created dataset which matches election day rainfall to electoral statistics across constituencies in Ireland, we show that rainfall has a significant negative effect on voter turnout. Our results indicate that a rainy day, defined as 30 mm of rain in 24 h, reduces voter turnout by approximately two percentage points (or three percent).

Using a theoretical framework based on a rational voting model, we show how population density may interact with rainfall to impact voter turnout. We propose two possible channels. Firstly, voters in rural areas may have a stronger sense of civic duty, compared to urban voters,

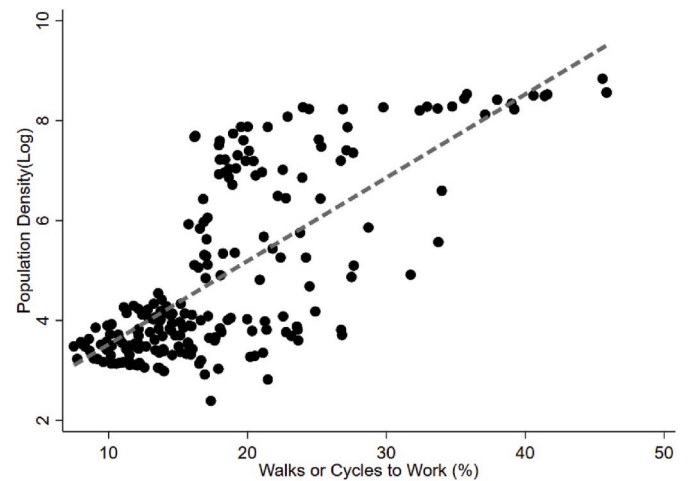


Fig. 6. Population density and the percentage of people that travel to work on foot or bicycle by constituency, 1997–2016. Dashed line represents the fitted least squares regression line.

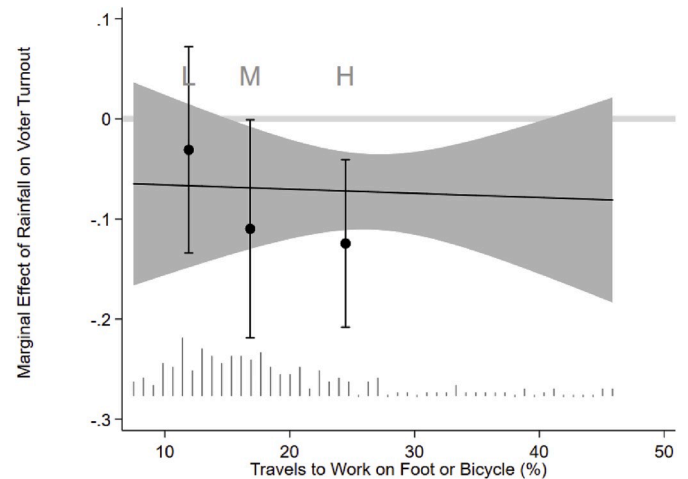


Fig. 7. The marginal effects of rainfall on voter turnout is shown across the mode of transport distribution, along with 95 percent confidence intervals. Three point estimates from the binning estimator show the effect of rainfall on voter turnout for low, medium and high percentages of people that travel by foot or bicycle.

which makes them immune to election day rainfall. Secondly, mode of transport may play a role. In urban areas, people are more likely to travel by foot or bicycle, whereas in rural areas travel by car is more common. Therefore, the effect of rainfall on turnout may be greater for urban voters.

In our empirical analysis, we detect the presence of interaction effects which show that the impact of rain on turnout increases as population density increases. In constituencies with a high population density, a rainy day is associated with a reduction in turnout of three percentage points (or five percent). We investigate the potential role of mode of transport using constituency level data on the percentage of people that travel to work on foot or by bicycle. Our analysis provides some empirical support to our hypothesis that mode of transport may interact with rainfall to impact turnout. However, given that we are using a proxy measure of how people may travel to the polling station, we cannot make definitive causal claims.

We also examine the role of a number of other explanatory variables. Our results show that the average age in a constituency is an important determinant of turnout. A one year increase in average age is associated

with an increase in turnout of between one and two percentage points. Age is also shown to interact with rainfall to influence turnout. Rain impacts turnout only in constituencies with young populations. In constituencies with older populations, there is no effect. This is consistent with age acting as a proxy for civic duty. Educational attainment, marriage rates and electoral uncertainty are also found to be positively associated with voter participation.

While it may not be possible to plan elections to coincide with good weather, there are policies which could potentially mitigate the negative effects of rainfall. For example, elections could be held at a time of year during which rainfall amounts are known to typically be low. Furthermore, elections could be held consistently at weekends. This would likely give voters a greater choice in terms of choosing a time to travel to the polling station in an attempt to avoid bad weather. The use of free public transport on election days, which has already been pioneered in US cities such as Los Angeles, Dallas, Houston and Tampa, may incentivise constituents, particularly those in urban areas, to vote. In Estonia, voters can cast their vote online in an early voting period several days before the election date. In Switzerland, all voters have the option of mailing in their vote in a three week period before the election date.

These types of policies could mitigate the impact of adverse weather on voter turnout.

Finally, the implications of our results may be particularly important in light of changing rainfall trends in Ireland which are attributed to climate change. According to the Irish Environmental Protection Agency (EPA), rainfall is increasing in northern and western areas of Ireland, while some areas of the south and east are experiencing decreases.¹³ If low turnout is more detrimental to certain parties, and if there is regional variation in party affiliation across Ireland, then changing rainfall trends may result in non-negligible impacts on electoral outcomes in Irish elections. While disentangling these effects is beyond the scope of the current paper, future research could examine these issues.

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Appendix

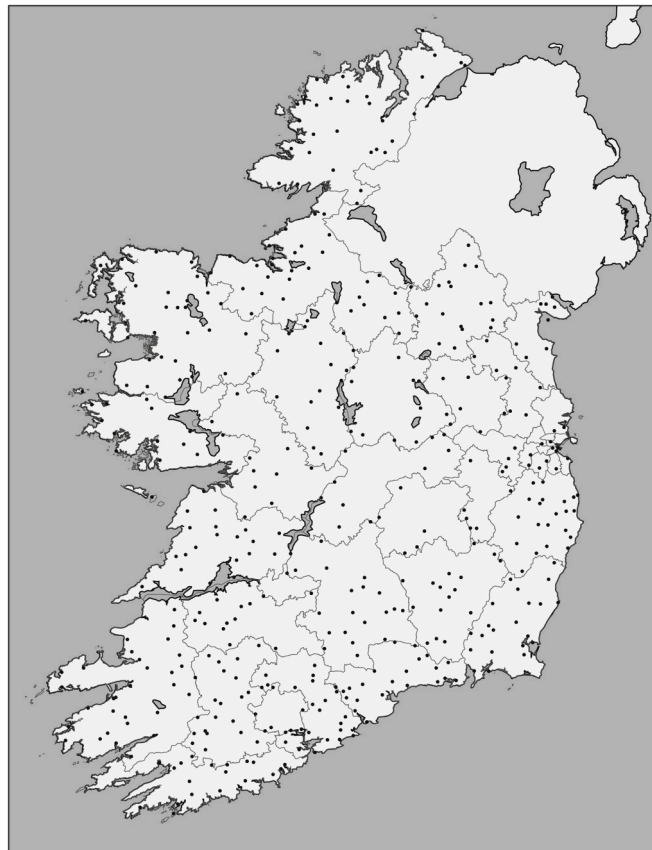


Fig. A.1. Met Éireann's weather stations are denoted by black dots. The lines on the map show the boundaries of the constituencies used in the 2016 election.

¹³ Further details are available on the EPA's website, visit www.epa.ie.

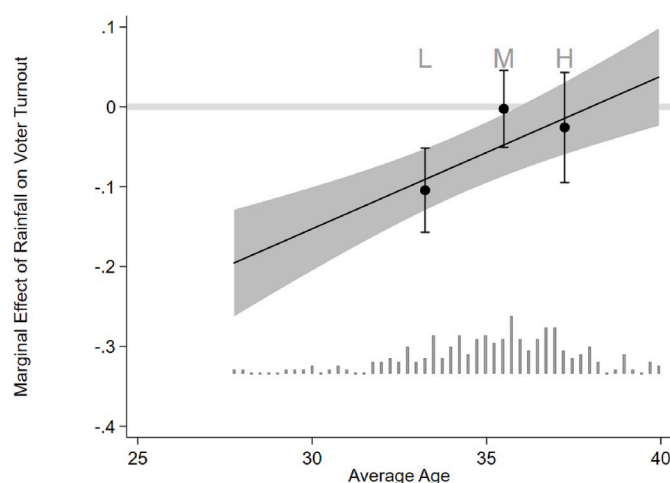


Fig. A.2. The marginal effect of rainfall on voter turnout is shown across the age distribution, along with 95 percent confidence intervals. Three point estimates from the binning estimator show the effect of rainfall on voter turnout for low, medium and high ages.

Table A1
Robustness of interaction effects

Interaction	Rain 1	Rain 2	Rain 3
Rainfall & Population Density	−0.036***	−0.036***	−0.043***
Rainfall & Age	0.018***	0.018***	0.020***

Notes: We test whether the population density and age interactions are robust to estimating Equation (8) with a full set of interactions for all independent variables. We do this for our baseline rainfall measure (“Rain 1”), our deviation from average measure (“Rain 2”) and our population weighted average measure (“Rain 3”). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.electstud.2020.102128>.

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